

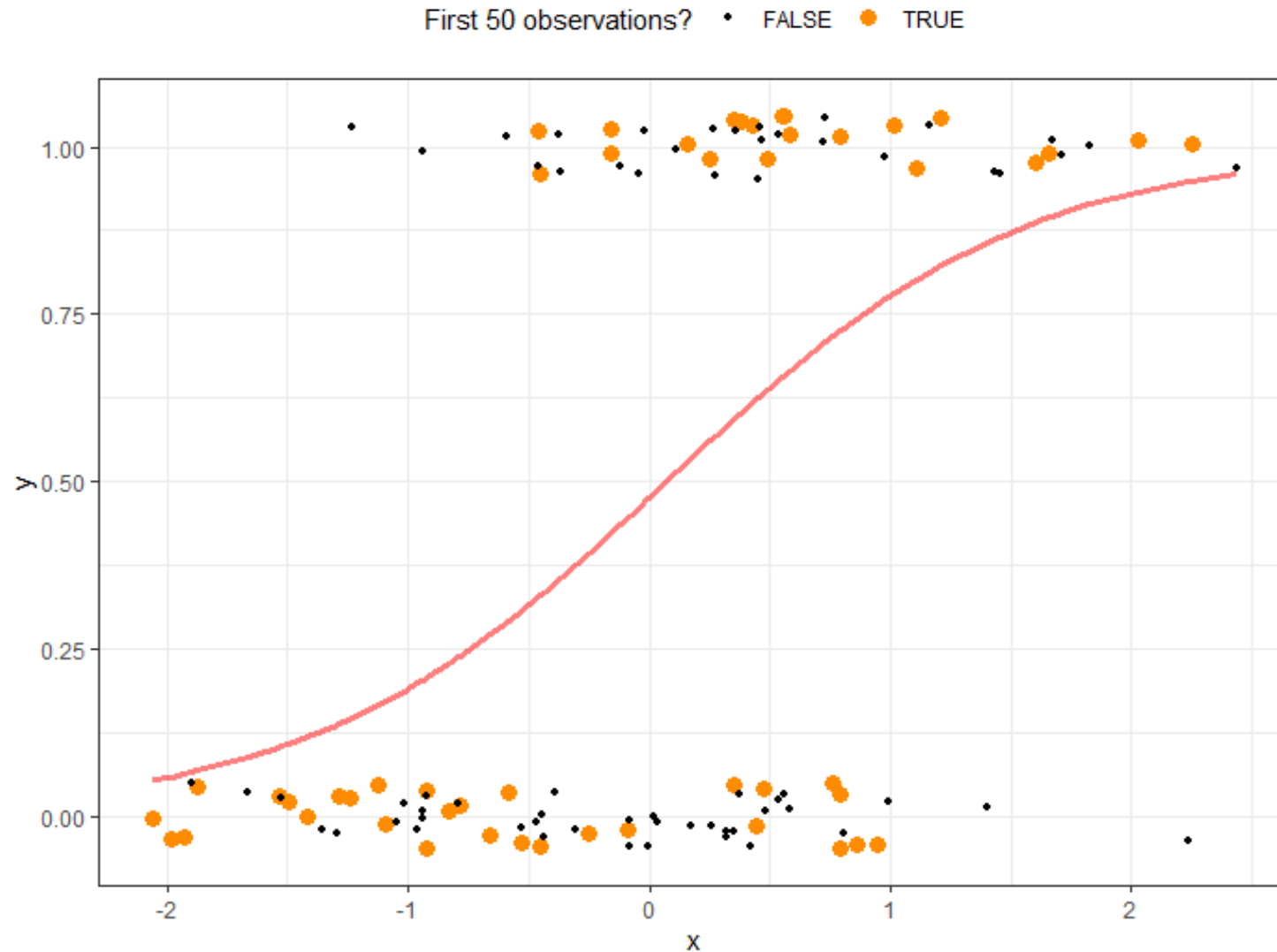
Continue to use the same single input linear relationship model we have focused on

$$\eta_n = \beta_0 + \beta_1 x_{n,1}$$

- Data were generated by specifying true parameter values of:

$$\beta_0 = -0.1 \text{ and } \beta_1 = 1.35$$

We will train a logistic regression model based on the first 50 observations



Use the Laplace Approximation to fit the Bayesian logistic regression model

- Use independent Gaussian priors with a shared prior mean and a shared prior standard deviation.
- Use a weak or diffuse prior with a prior standard deviation of 20 for both parameters.

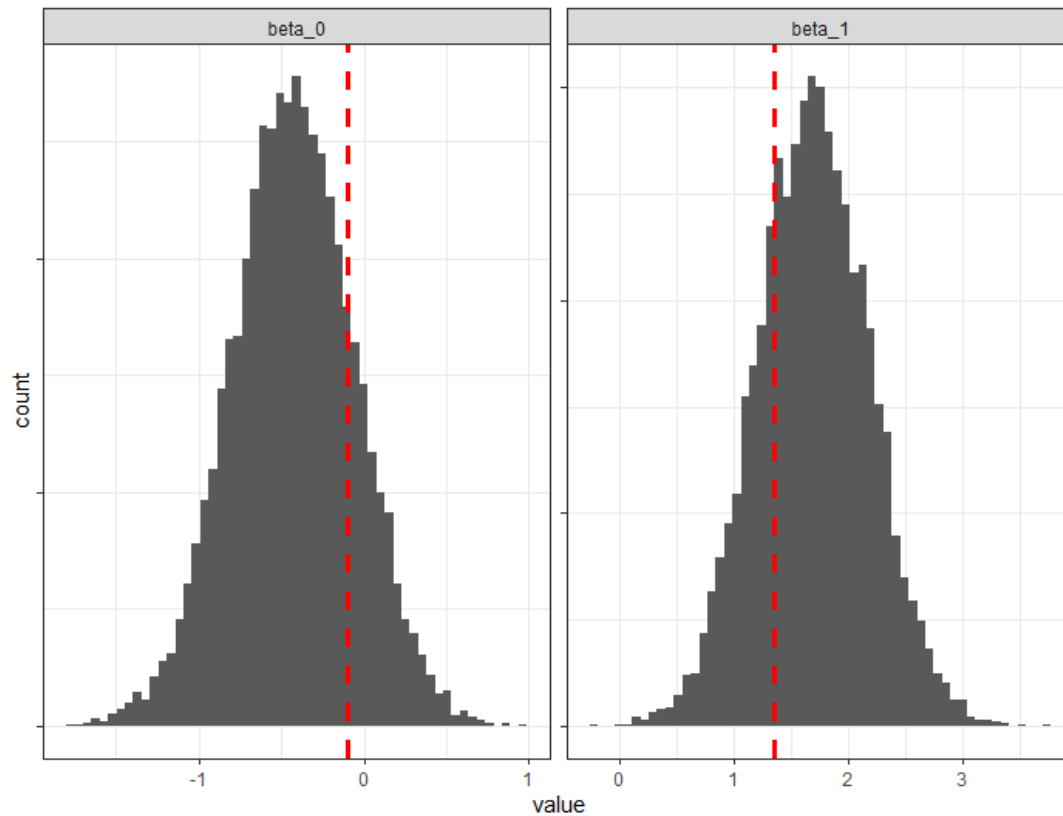
$$\boldsymbol{\beta} \mid \mu_{\beta}, \tau_{\beta} \sim \prod_{d=0}^D \left(\text{normal}(\beta_d \mid \mu_{\beta} = 0, \tau_{\beta} = 20) \right)$$

Execute the Laplace Approximation

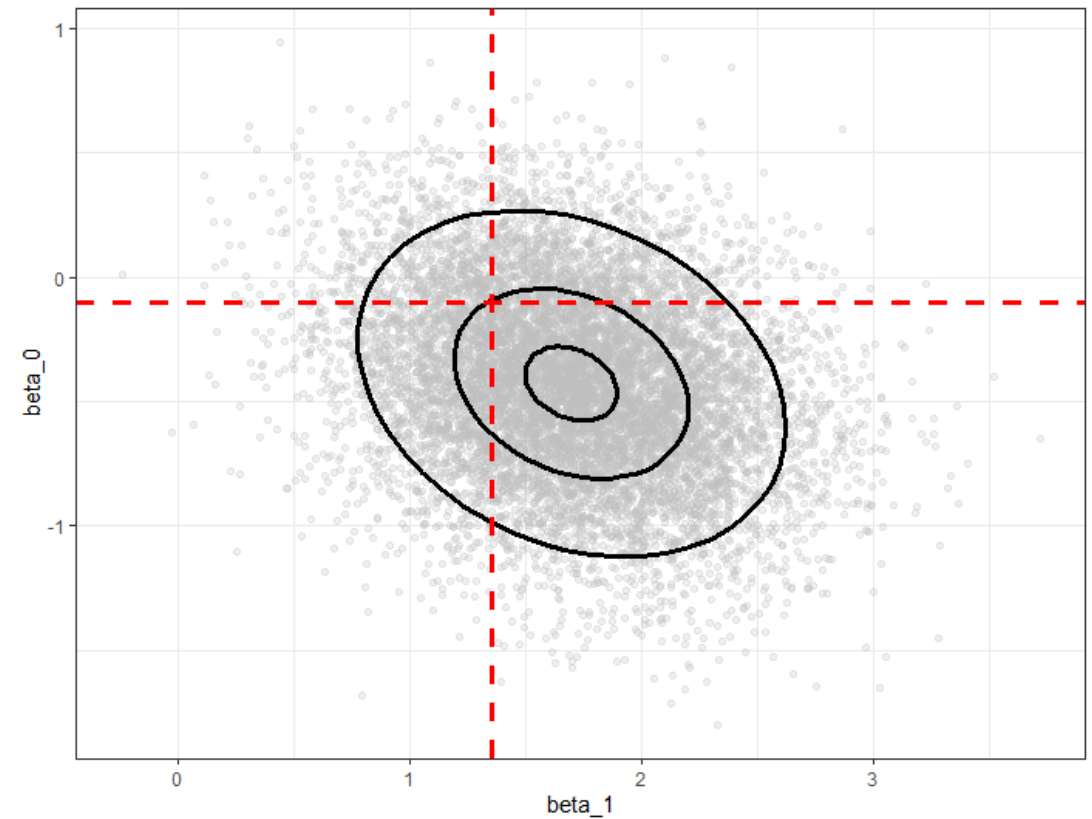
- Setup the log-posterior function.
- Specify the training set design matrix.
- Package the design matrix, training response, prior specification into an appropriate list.
- Run the Laplace Approximation.

Inspect the posterior distributions on the parameters

Marginal posterior histograms



posterior scatter plot

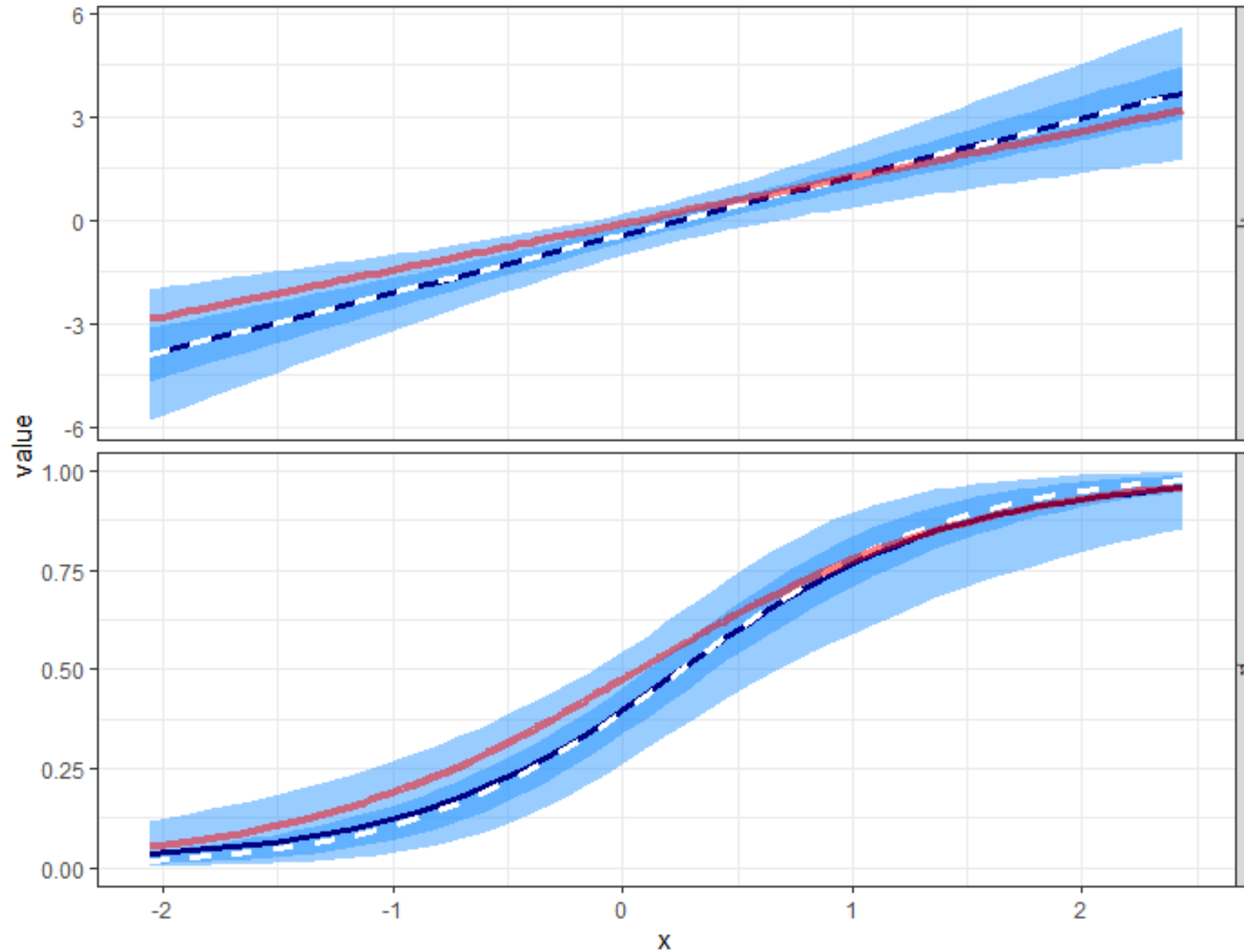


Posterior correlation coefficient: -0.23

How do we make posterior predictions?

- Same setup as with linear models!
- Generate posterior samples of the β -parameters.
- Calculate the linear predictor for each posterior sample.
- Calculate the probability via the logistic function for each posterior sample.

Posterior predictions over a test set for visualization purposes



Logistic regression predicts a probability

- We do not observe a probability...we observe a binary outcome where the response takes on one of two classes, $y = \{0,1\}$.
- How do we convert the predicted probability to a predicted classification?

Make decisions!

- As discussed earlier in the semester, we classify by comparing the predicted probability to a threshold value.
- The threshold is commonly set to 0.5.
- If the predicted probability is greater than the threshold we classify the outcome as the EVENT.

Compare the predicted classes to the observed classes to determine classification performance metrics

- Point wise predicted accuracy at a single threshold:
 - Accuracy
 - Confusion matrix – Sensitivity and Specificity
- Point wise predicted accuracy at many threshold values:
 - ROC curve – Area Under the ROC Curve (AUC)
- “Long term consistency” or well-calibrated models:
 - Calibration curves – graphical tool
 - Brier score for a single metric

Compare the predicted classes to the observed classes to determine classification performance metrics

- Point wise predicted accuracy at a single threshold:

- Accuracy
- Confusion matrix – Sensitivity and Specificity

We will focus on the confusion matrix in this presentation.

- Point wise predicted accuracy at many threshold values:

- ROC curve – Area Under the ROC Curve (AUC)

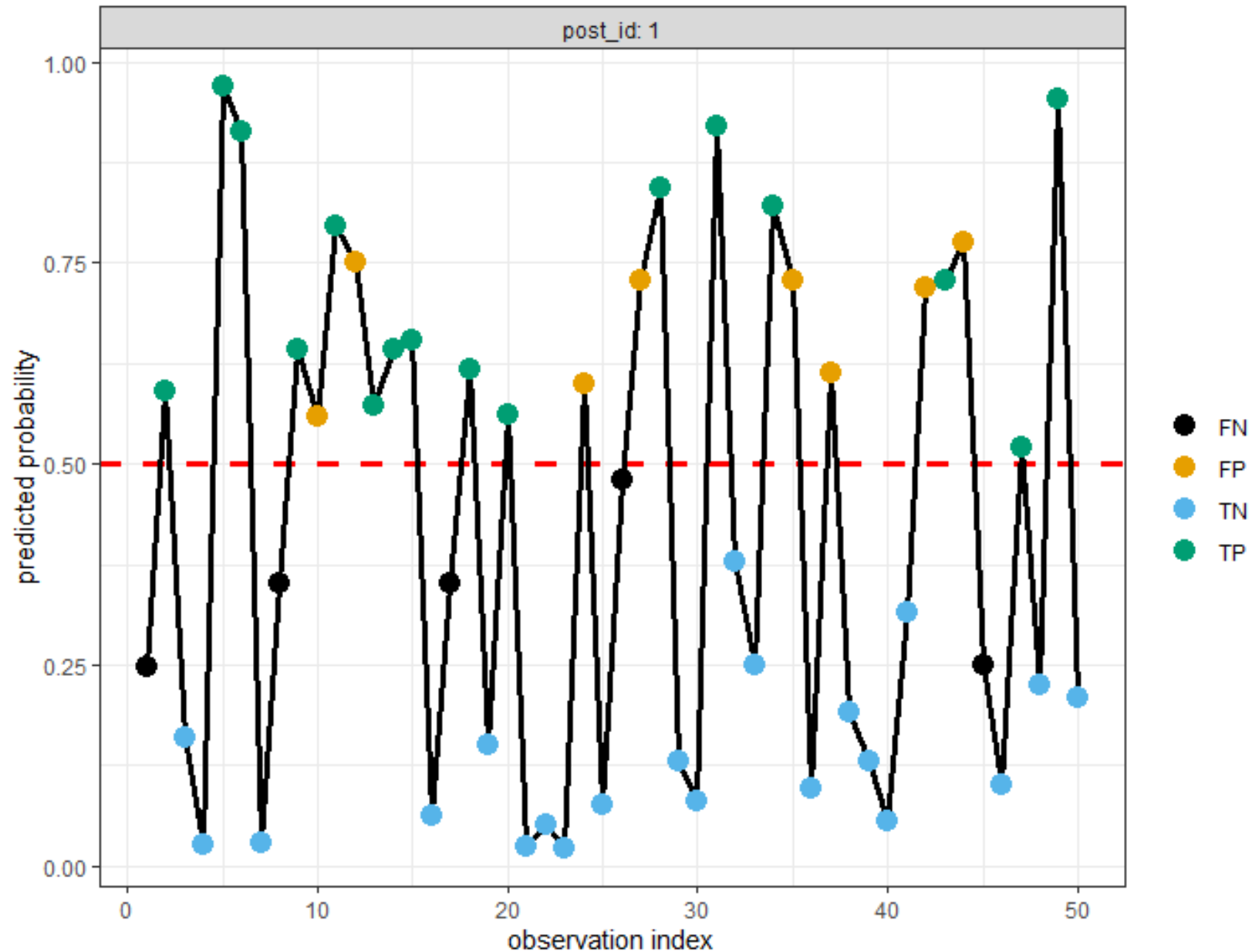
- “Long term consistency” or well-calibrated models:

- Calibration curves – graphical tool
- Brier score for a single metric

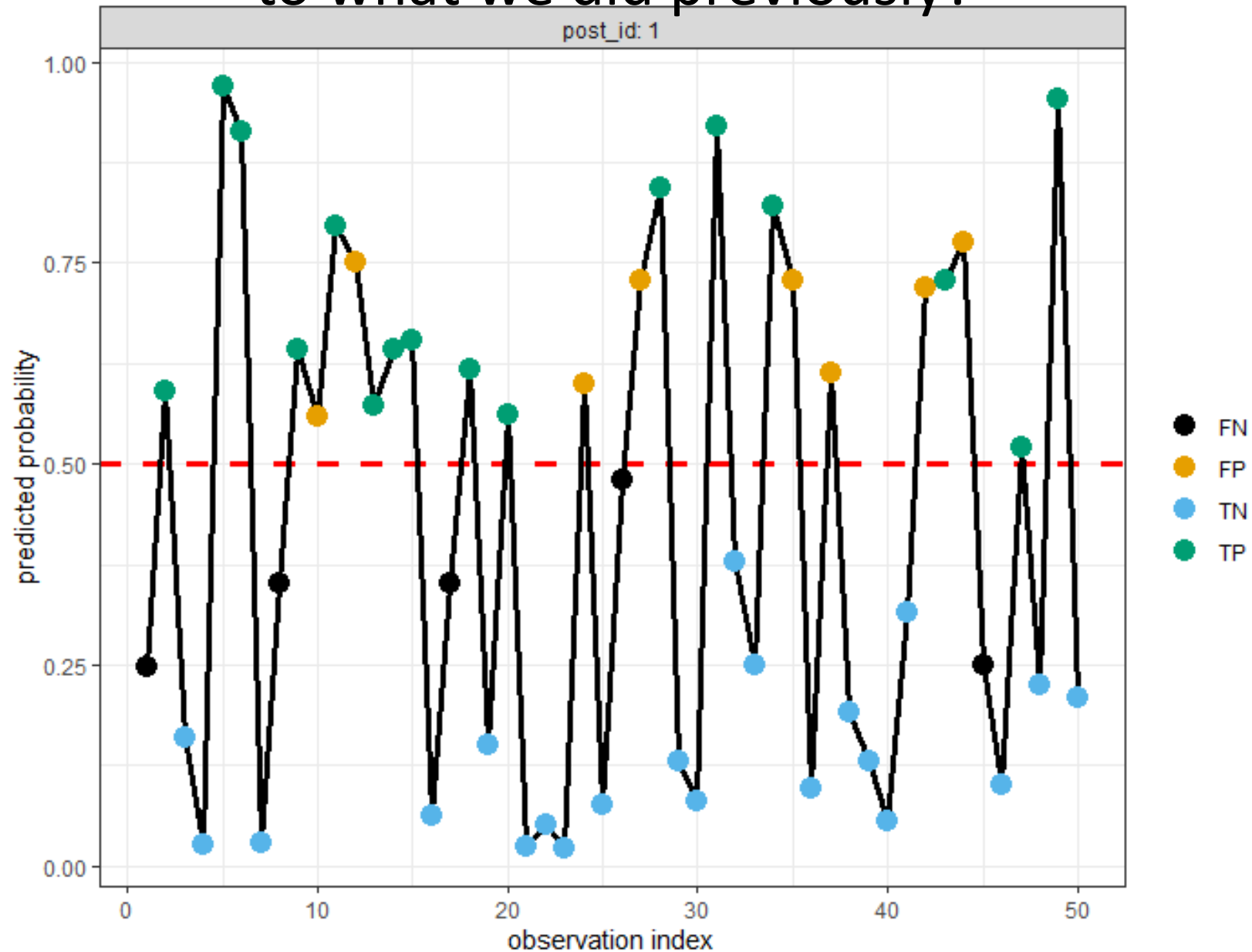
Predict and classify the training set, then evaluate performance

- Generate posterior samples for the parameters.
- Predict the training set.
- Set the threshold to be 0.5 and classify each prediction.
- After classification, compare the predicted class to the observed class.
- Calculate the training set confusion matrix.

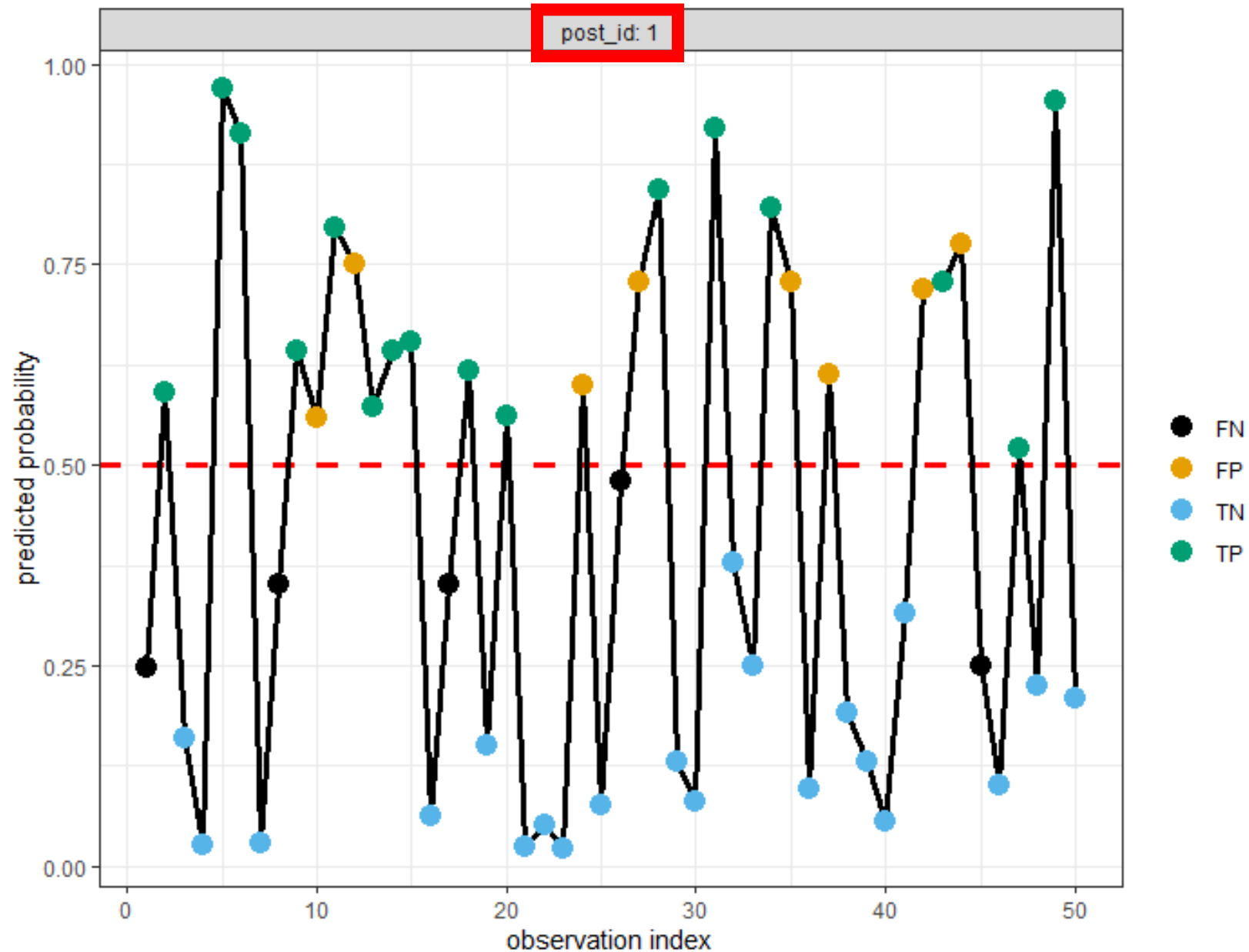
Classify with a 50% threshold value



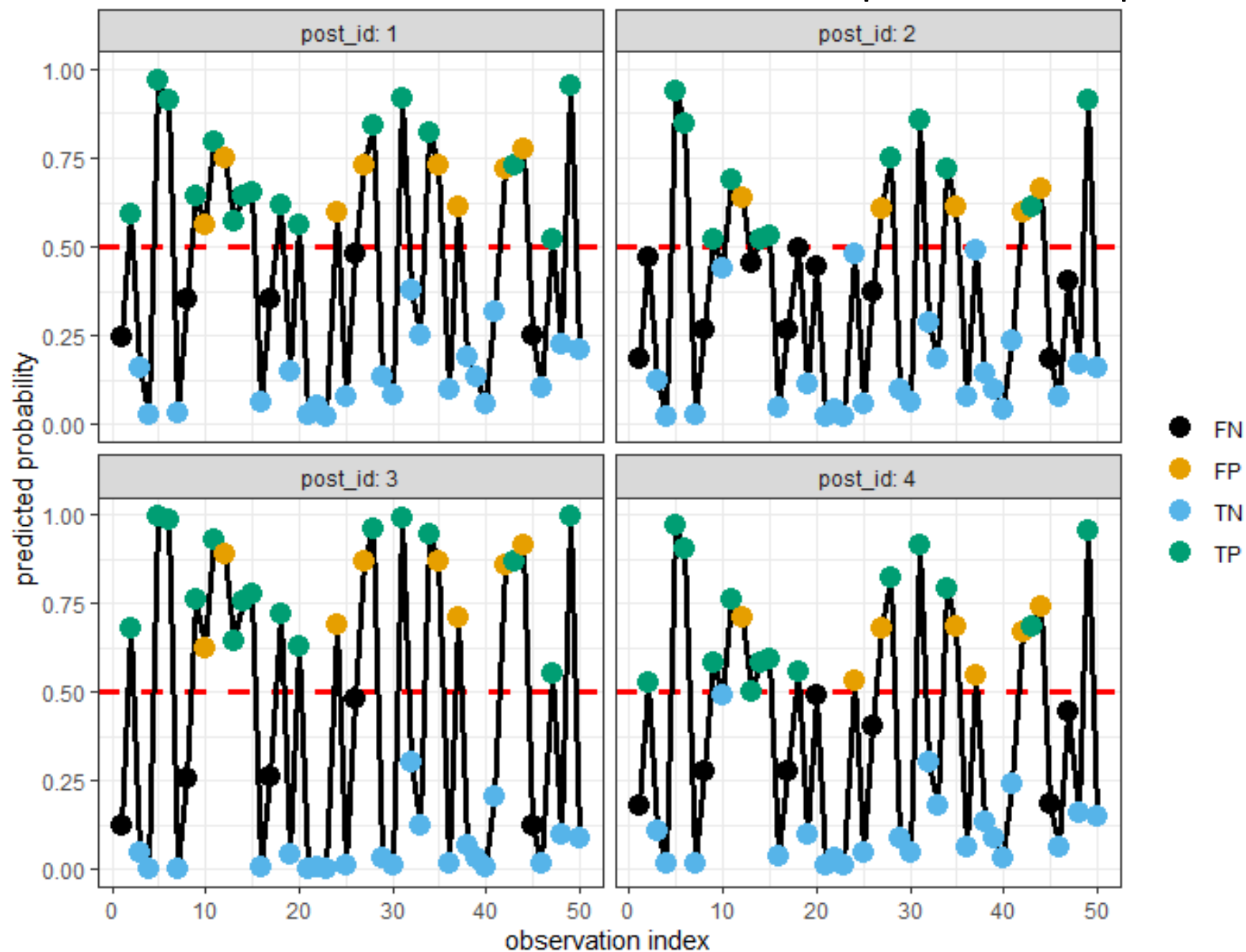
Is there anything different in the Bayesian setting compared to what we did previously?



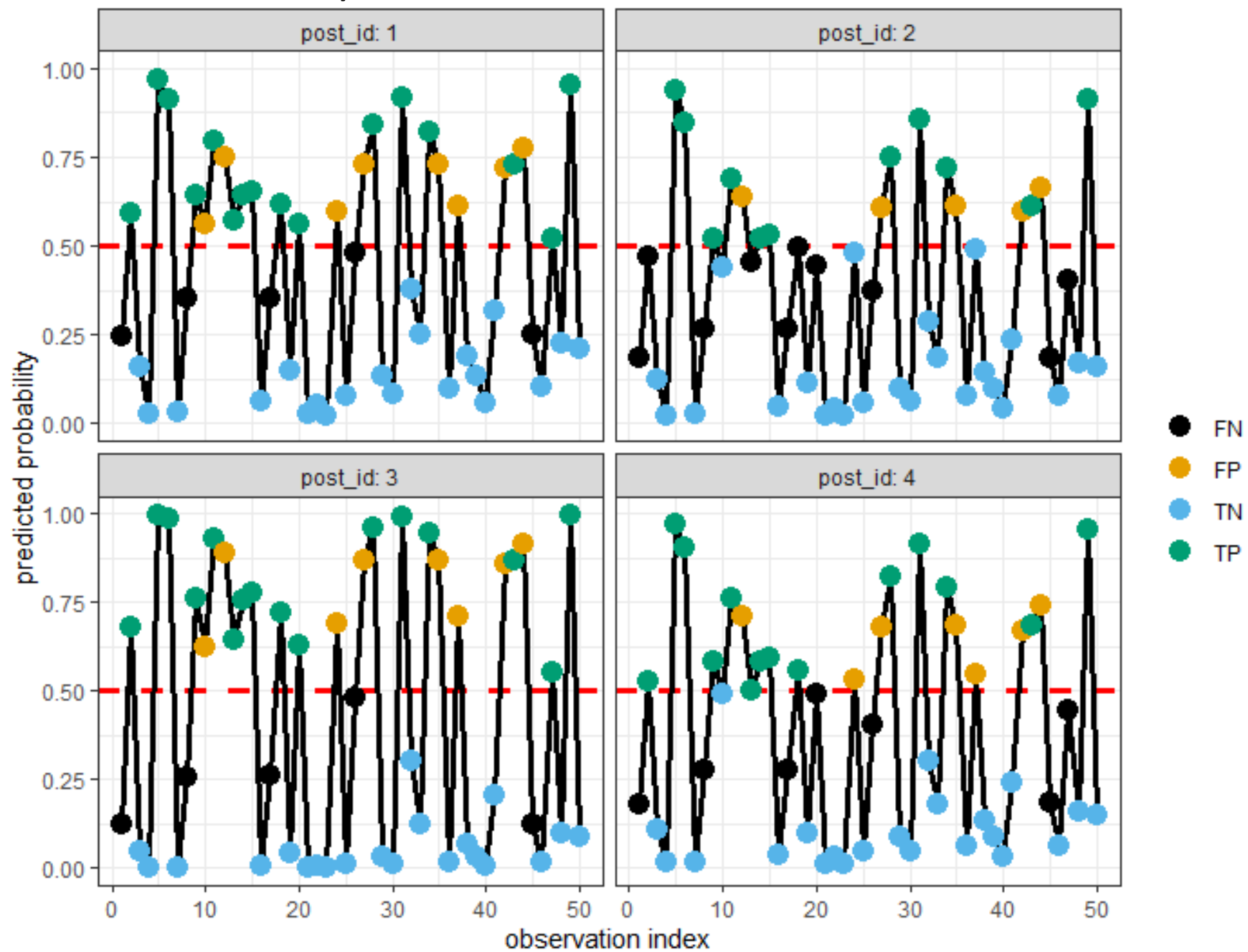
In a Bayesian setting, we make predictions for every posterior sample!



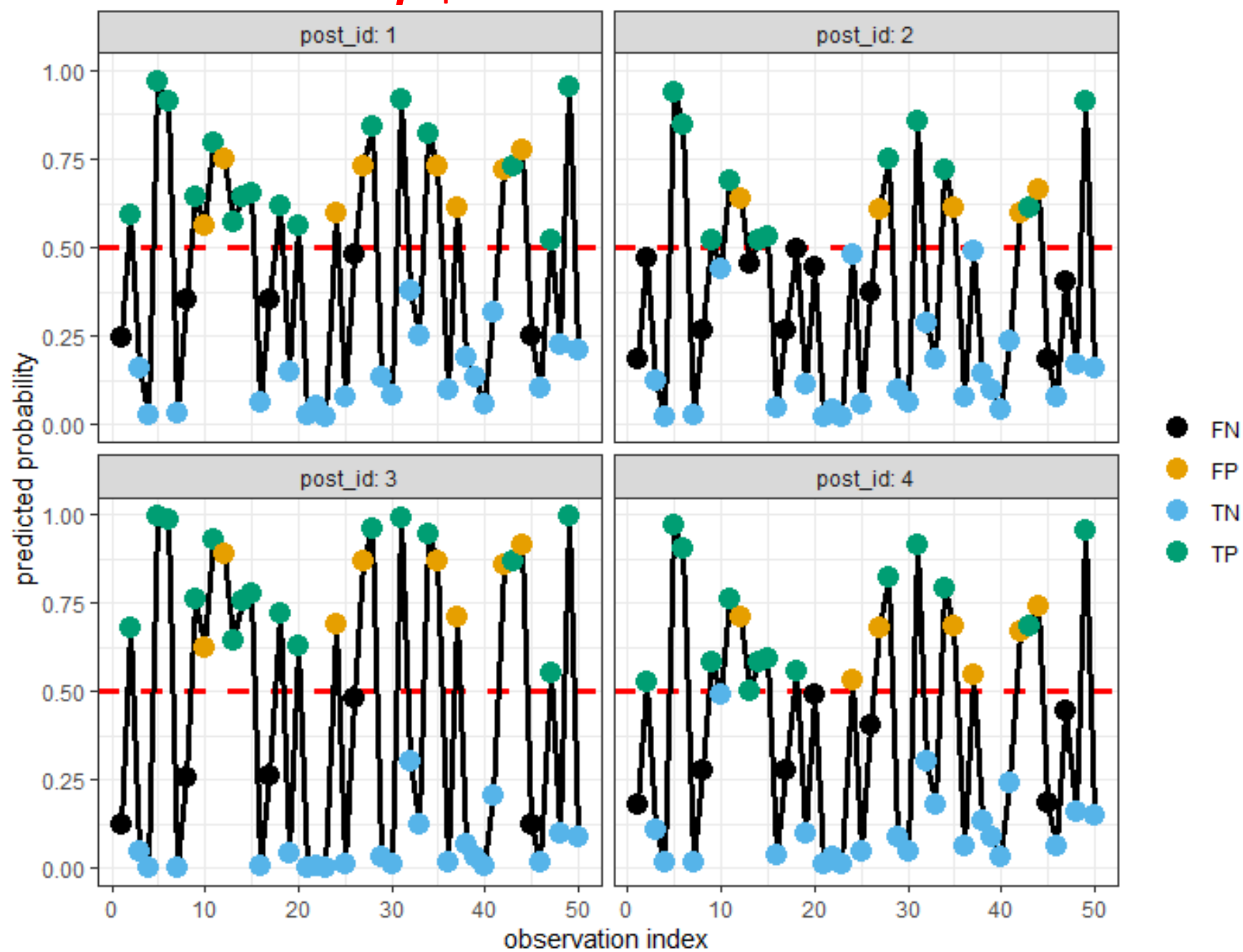
Predicted classifications based on the first 4 posterior samples



Why are the classifications different?

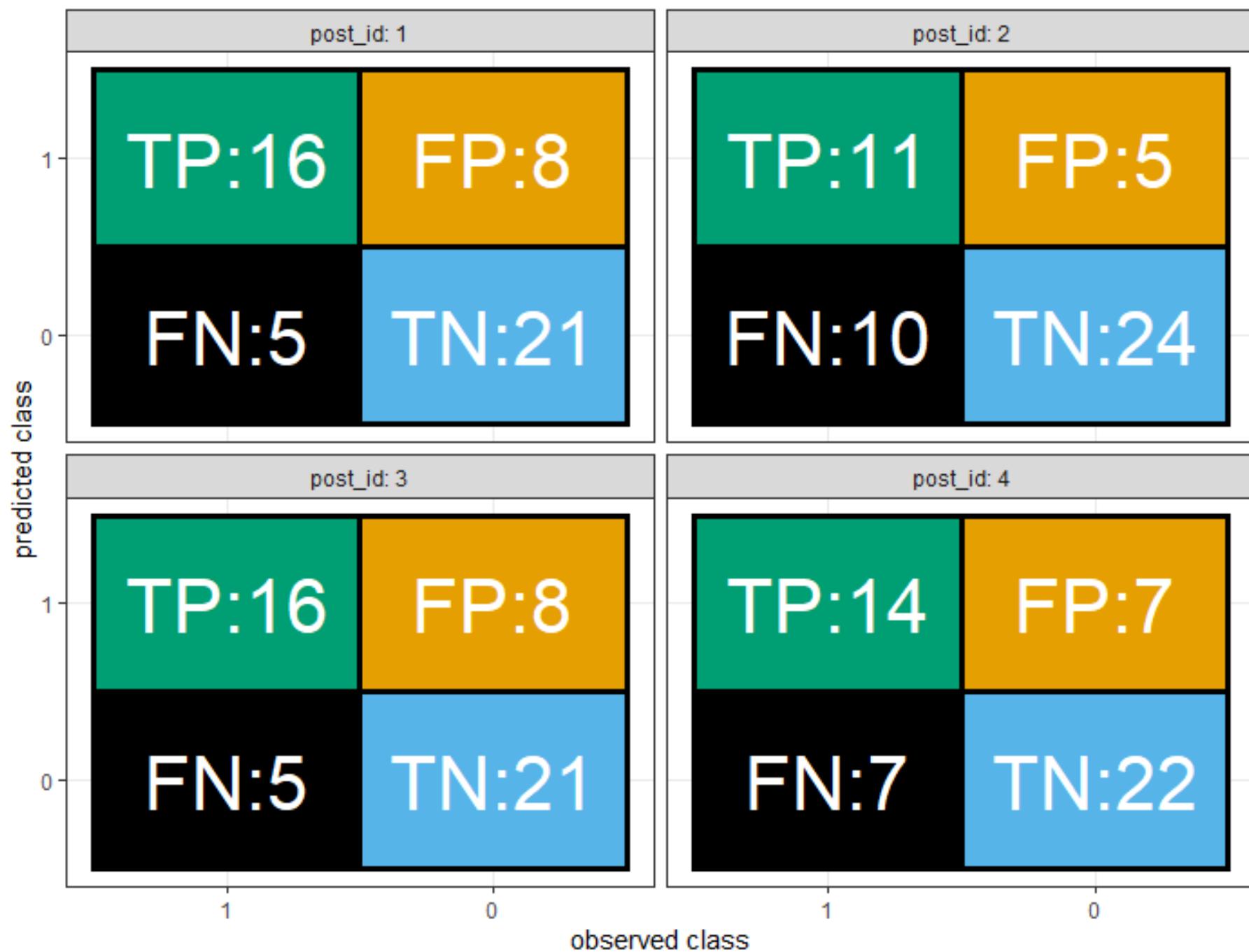


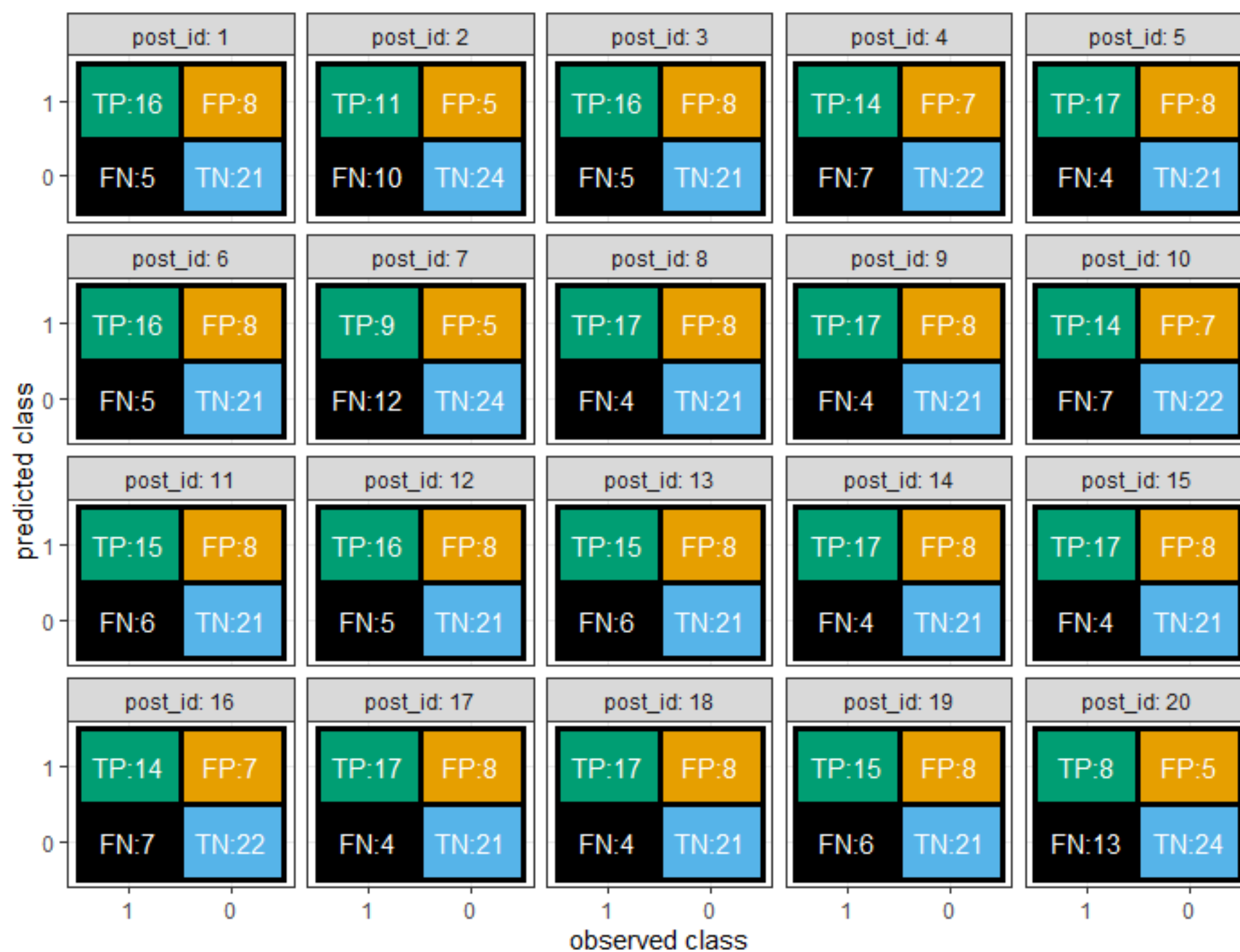
The β -parameters are different!



Count the observations associated with each element of the confusion matrix

- Count all True-Positives, True-Negatives, False-Positives, and False-Negatives.
- Perform this operation for each posterior sample!

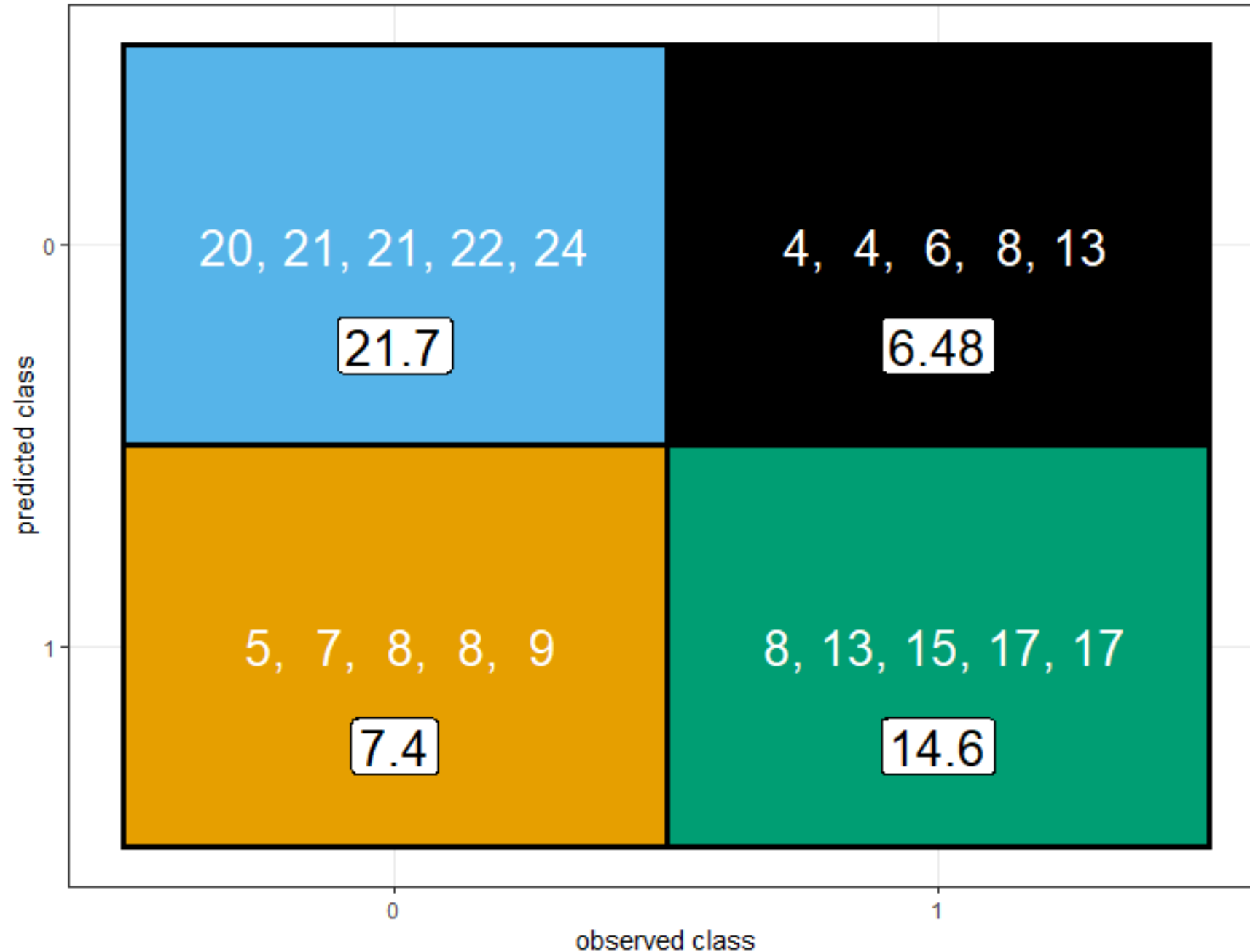




The different posterior predictions represent the model's UNCERTAINTY

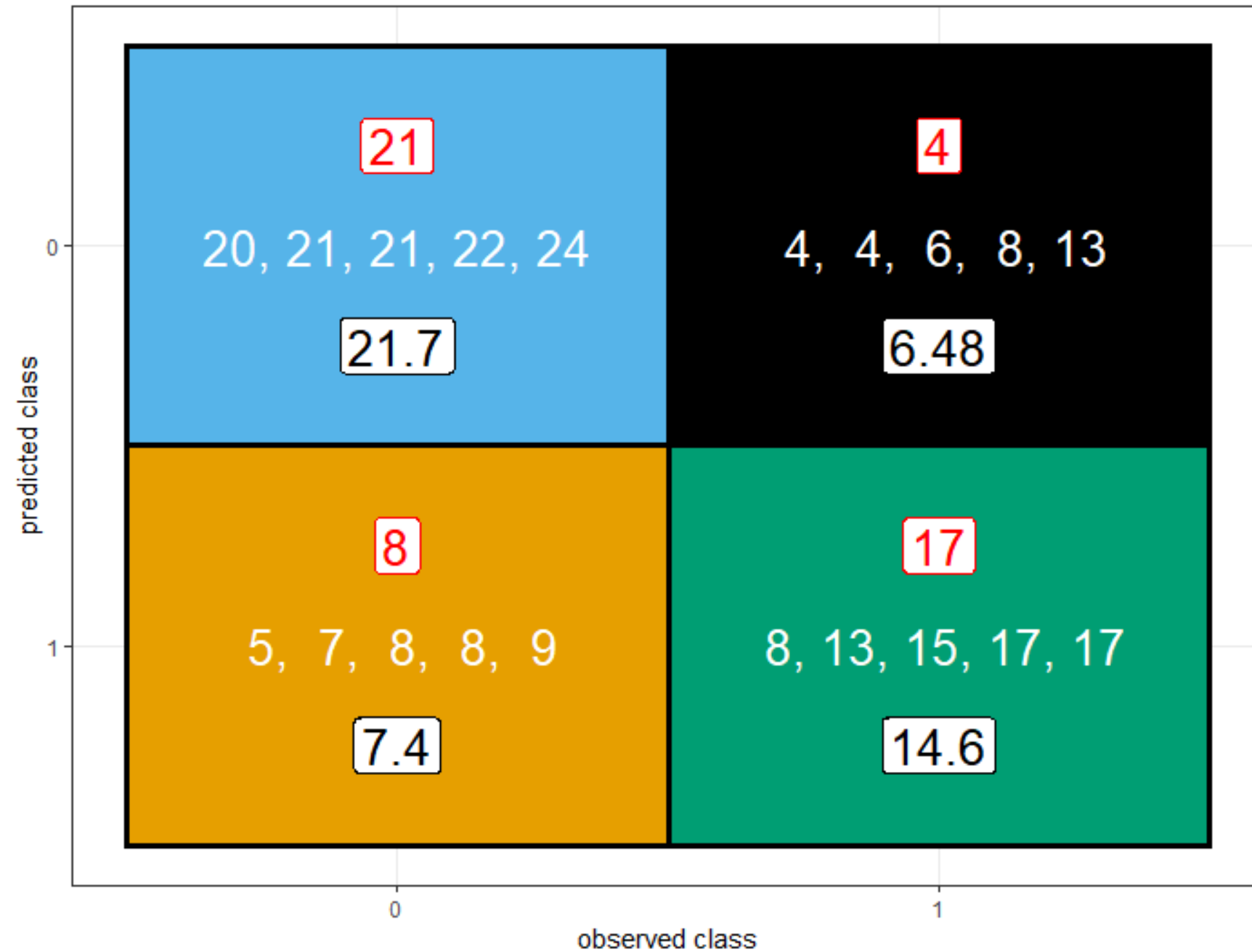
- We can summarize across all posterior predictions to assess the mean confusion matrix.
- We can also summarize uncertainty intervals on each of the counts in the confusion matrix.

Each confusion cell is summarized by the 5th, 25th, 50th, 75th, and 95th quantiles. Means are displayed in white text boxes.

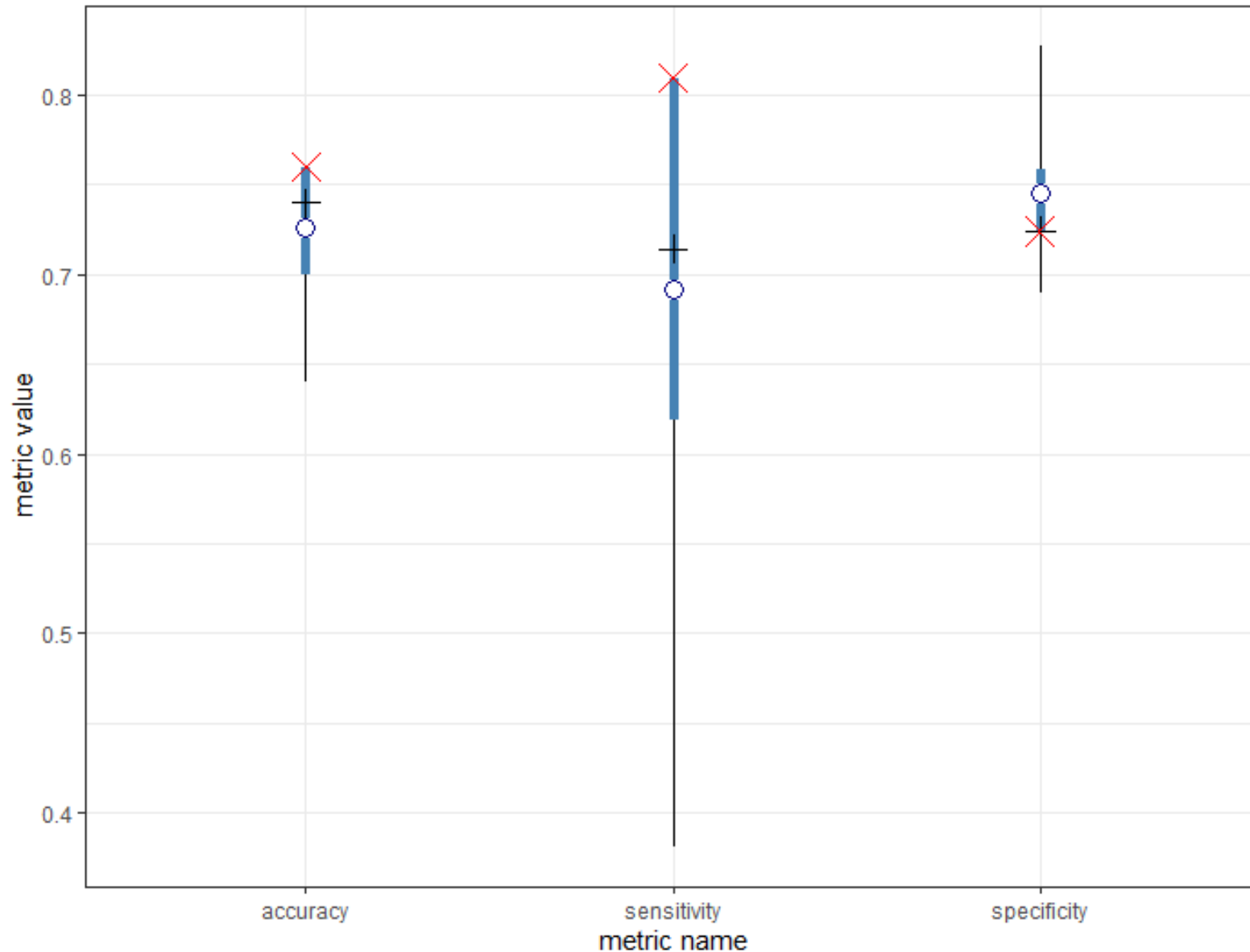


How well did we do on the training set?

- Since this is a toy demo we can calculate the TRUE confusion matrix and compare to the posterior predicted summary stats.



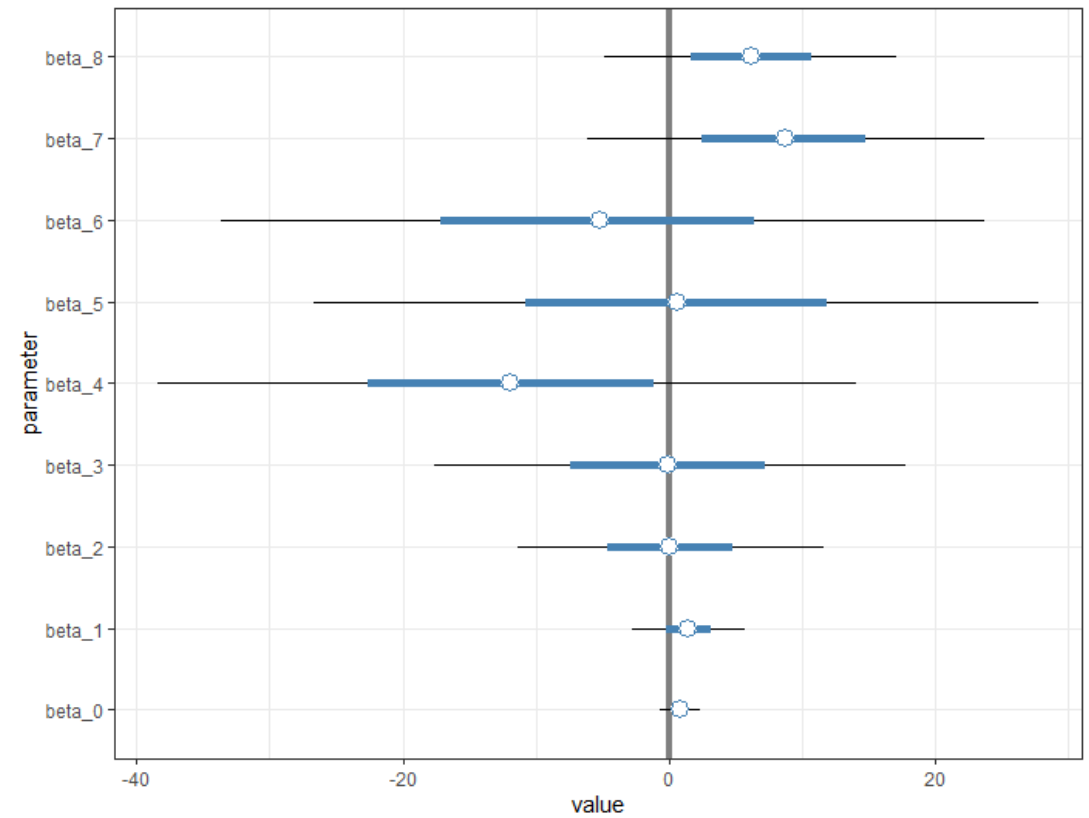
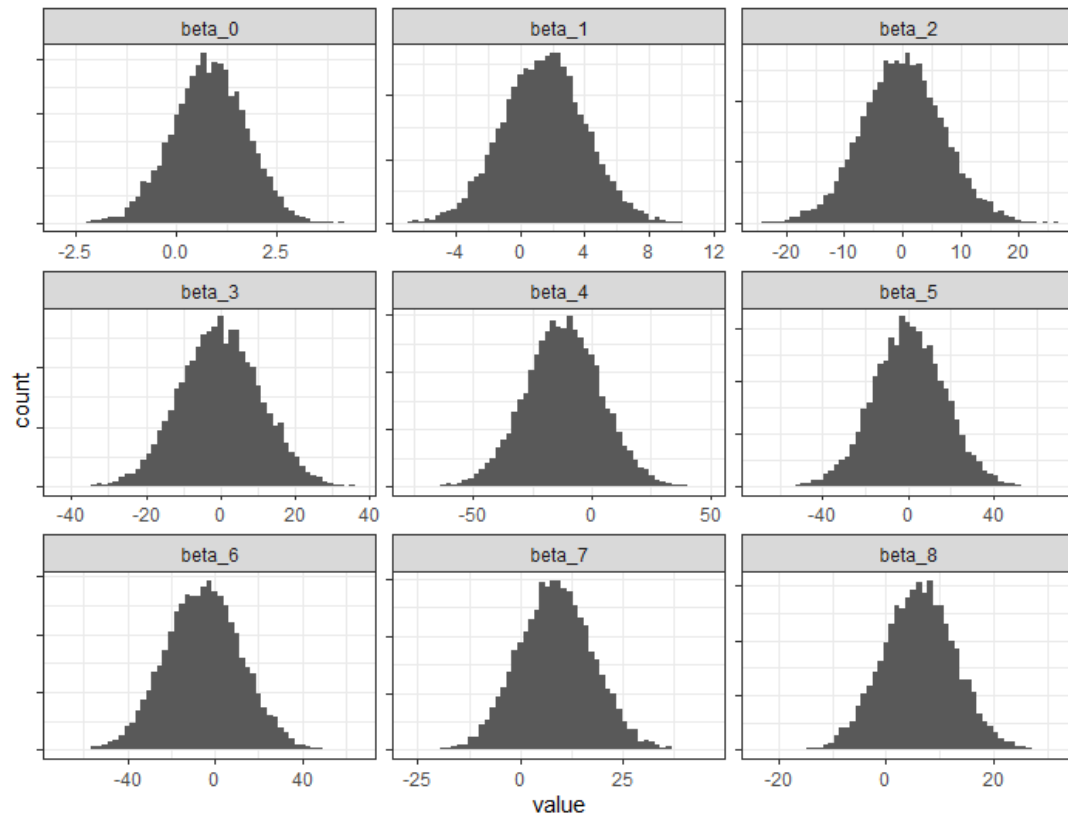
Calculate Accuracy, Sensitivity, and Specificity posterior summaries and compare with the TRUE answers



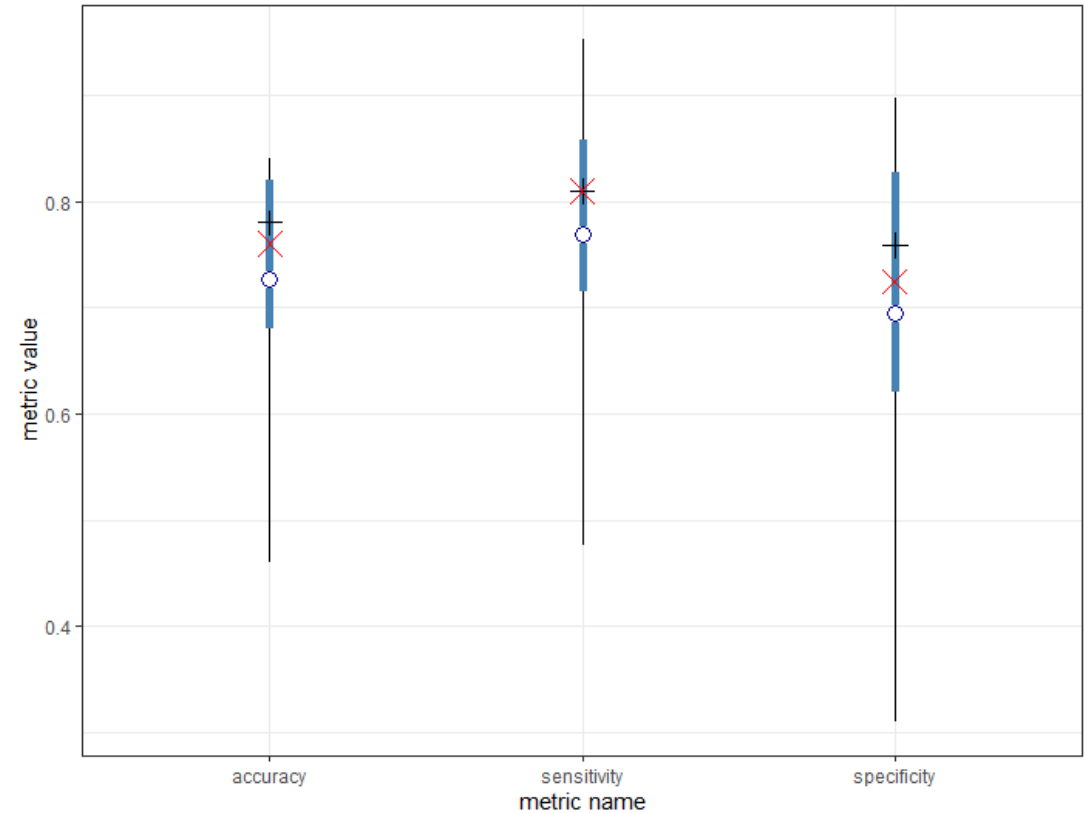
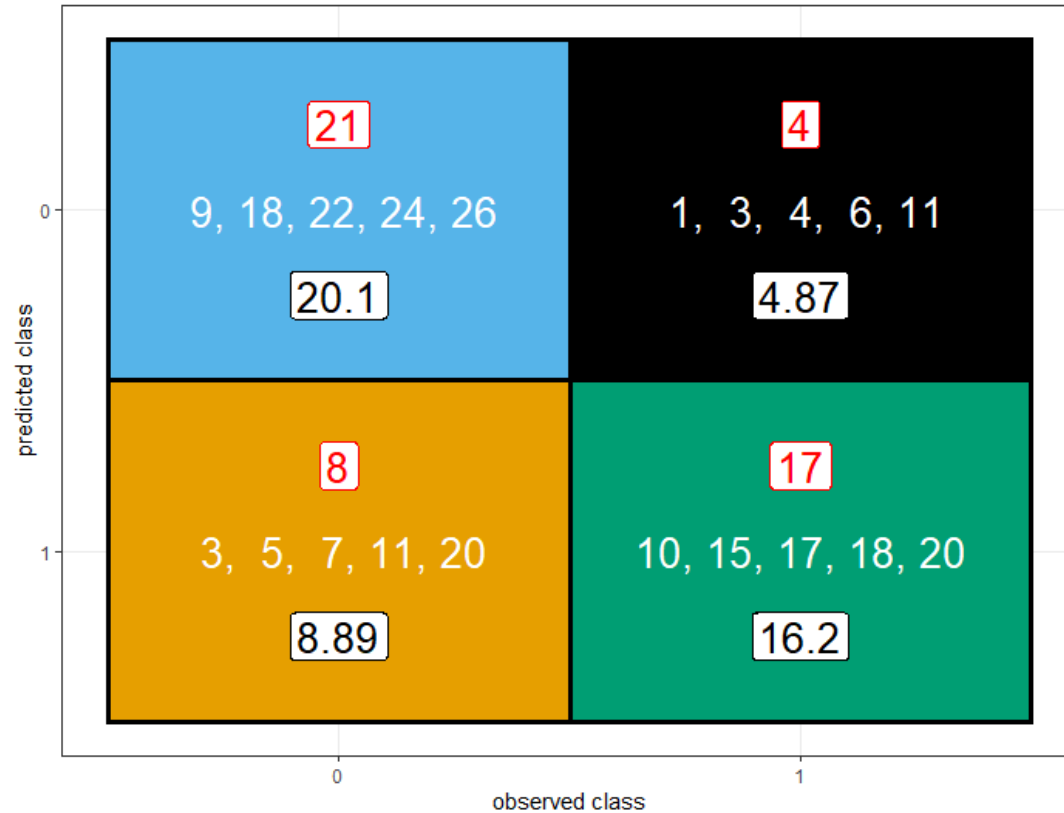
What would happen if we fit an 8th order polynomial to the data?

- Continue to use the weak or diffuse prior with a shared prior standard deviation of 20.
- Fit the Bayesian logistic regression model with the Laplace Approximation.

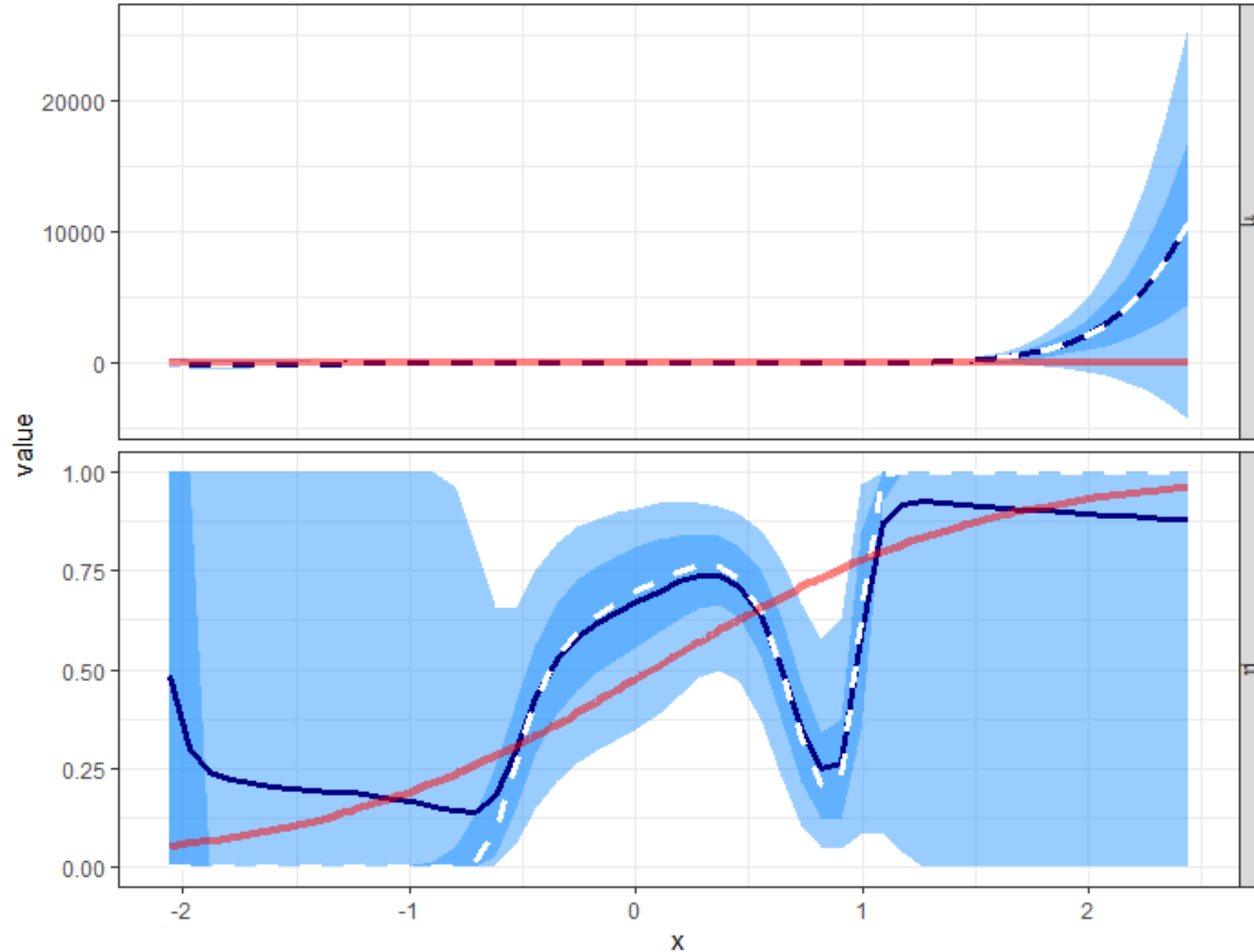
Inspect the posterior distribution



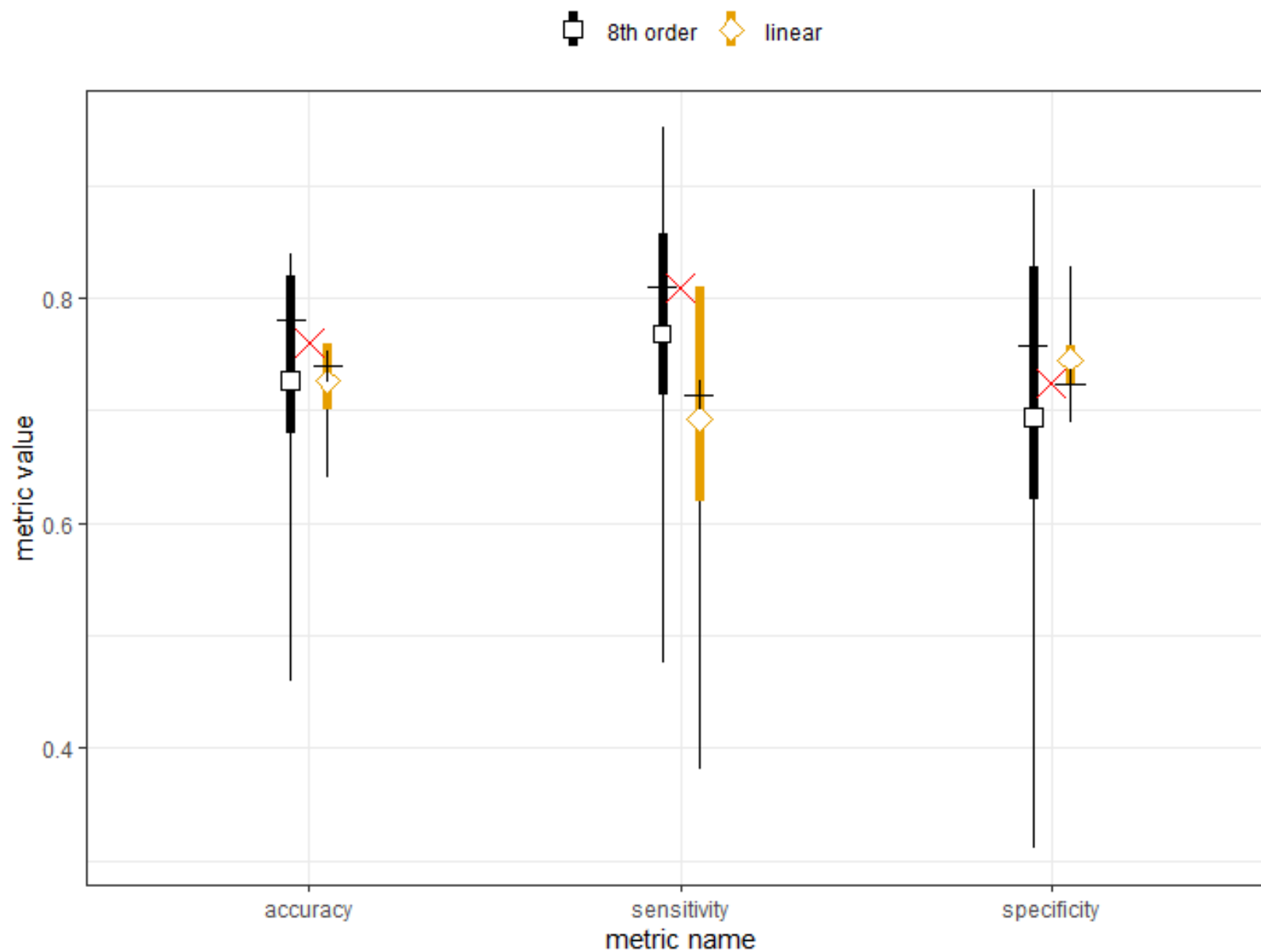
8th order polynomial training set performance



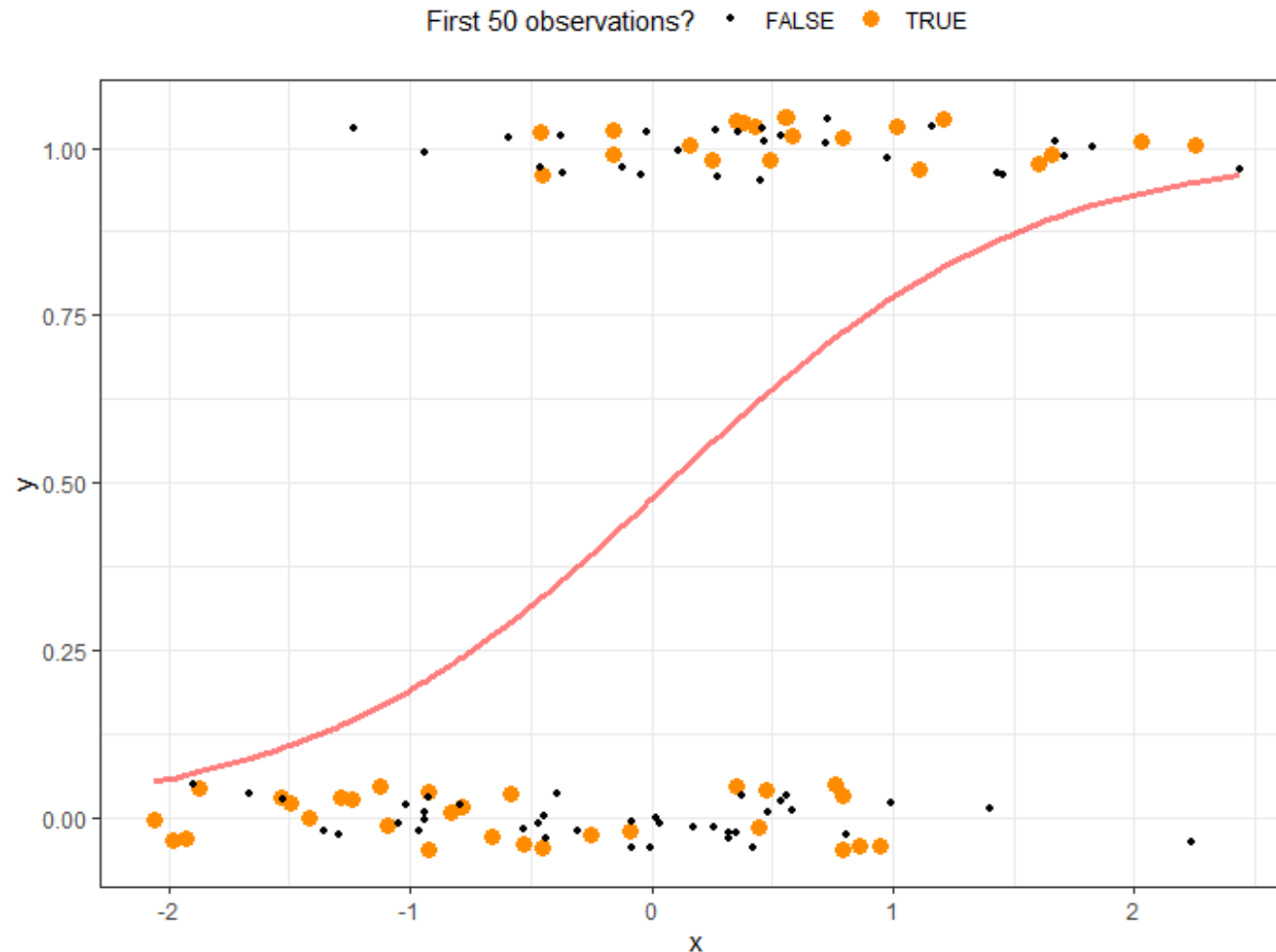
Visualize the posterior predicted trends...



Which model is better?



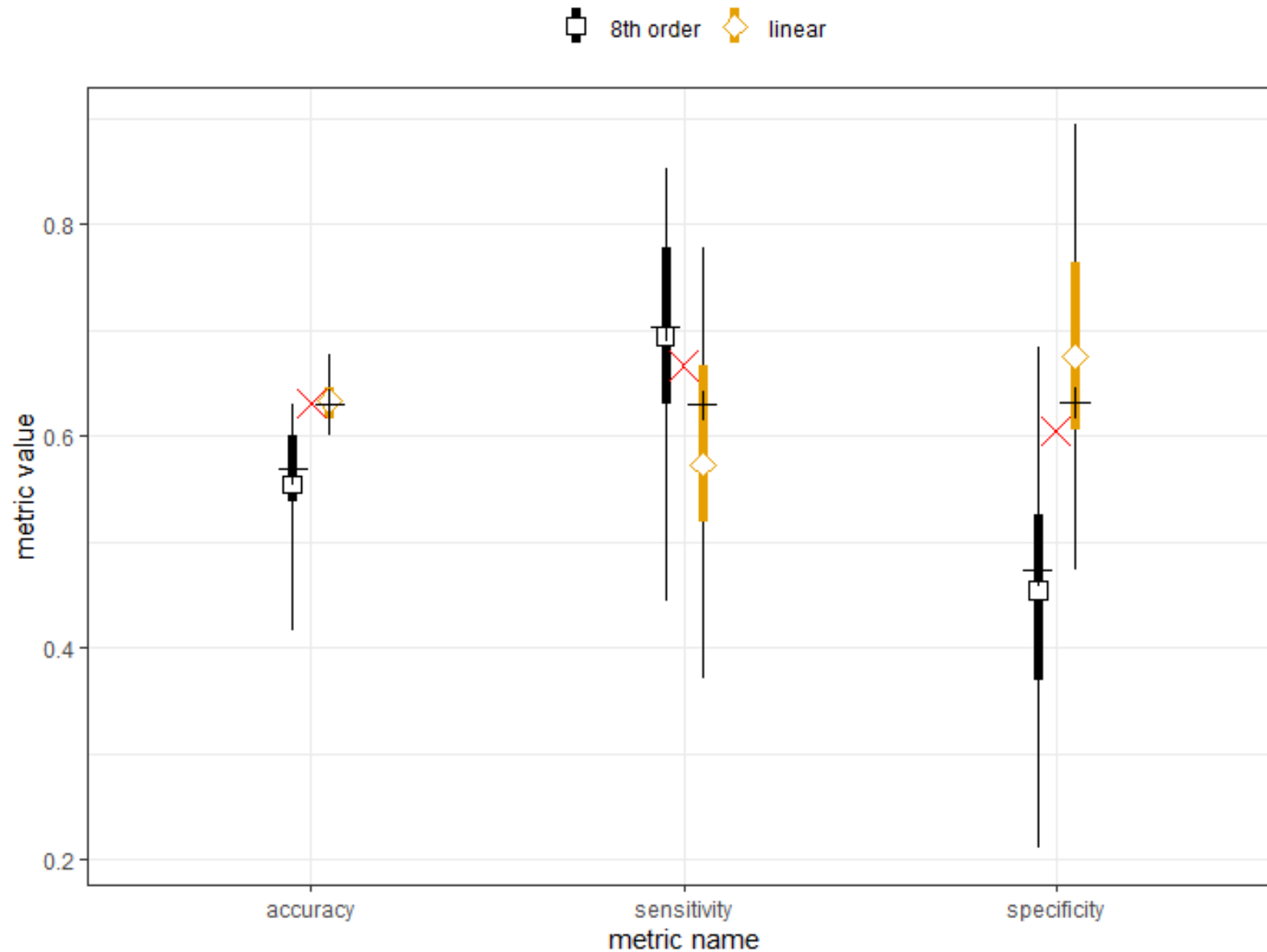
Let's consider the performance on the points we did not include in the training set



Predict the hold-out set

- Create the hold-out set design matrices.
- Make posterior predictions with all posterior samples.
- Calculate the confusion matrices associated with each posterior sample, for both models.
- Calculate Accuracy, Sensitivity, Specificity for each posterior sample, for both models.
- Summarize.

Which model is better?



We can also compare the models with the Laplace Approximation's estimate to the Evidence!

- Since we have two models we will calculate the Bayes Factor of the linear model relative to the 8th order polynomial.

$$\exp(\log[p(\mathbf{y} \mid \mathbf{X}, M_1)] - \log[p(\mathbf{y} \mid \mathbf{X}, M_8)]) \approx 7.25$$

- Strong evidence to support the linear model is better, based just off of the training set!!